

“I don't recognize this charge”

The least-resolved reason for calling a bank's contact center

1 · HER MOMENT

BANK · now · SYNTHETIC EXAMPLE

Charge alert

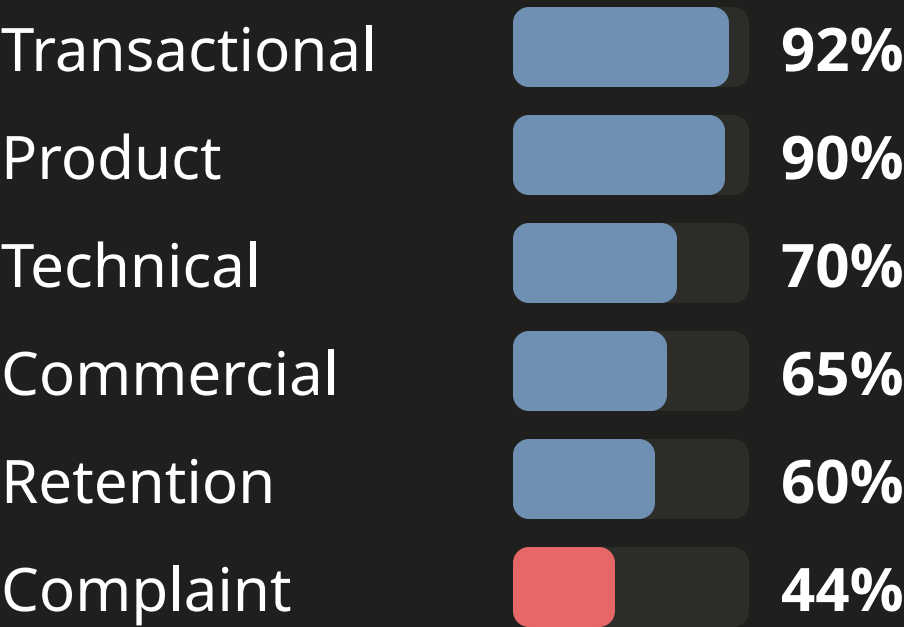
Purchase of US\$ 261.56 at
EMPRESA TELEFONICA with card
••3240.

Do you know this charge?

She cannot tell if it is hers, a
mistake, or fraud. She calls the
bank, and waits.

2 · WHAT THE BANK'S DATA SHOWS

Calls resolved, by reason



44% of complaint calls resolved · 431 s
median call duration · 63% of
complaint calls require follow-up

3 · WHY IT MATTERS

Customer satisfaction (scale 1–4)



1.0 point satisfaction gap

Resolved calls are associated with
higher satisfaction. We do not claim
cause.

97,851 surveys after resolved calls · 30,005
after unresolved

 the problem · not resolved  other reasons · resolved

12,297 of 67,095 complaints are tagged “unrecognized charge”. This is the case I chose.

Lora handles the case, or hands it to a person

Inside the bank's app, in Spanish and Portuguese.

"I don't recognize this charge"

In her own words
Spanish or Portuguese



Lora checks who she is, finds the matching transaction in her account, shows it and asks:
do you recognize it?



She recognizes the charge

The case closes. No claim.

She does not recognize it

She confirms → Lora opens the claim → re-reads the database → real case number and deadline.

Fraud signal above the certified cut-off

Lora offers to block the card; she decides. The case goes to a human fraud agent at priority one.

At any step she can ask for a person · the human agent receives the whole case · she repeats nothing

Out of scope today: deposits and disbursements → a person

For her

A clear answer

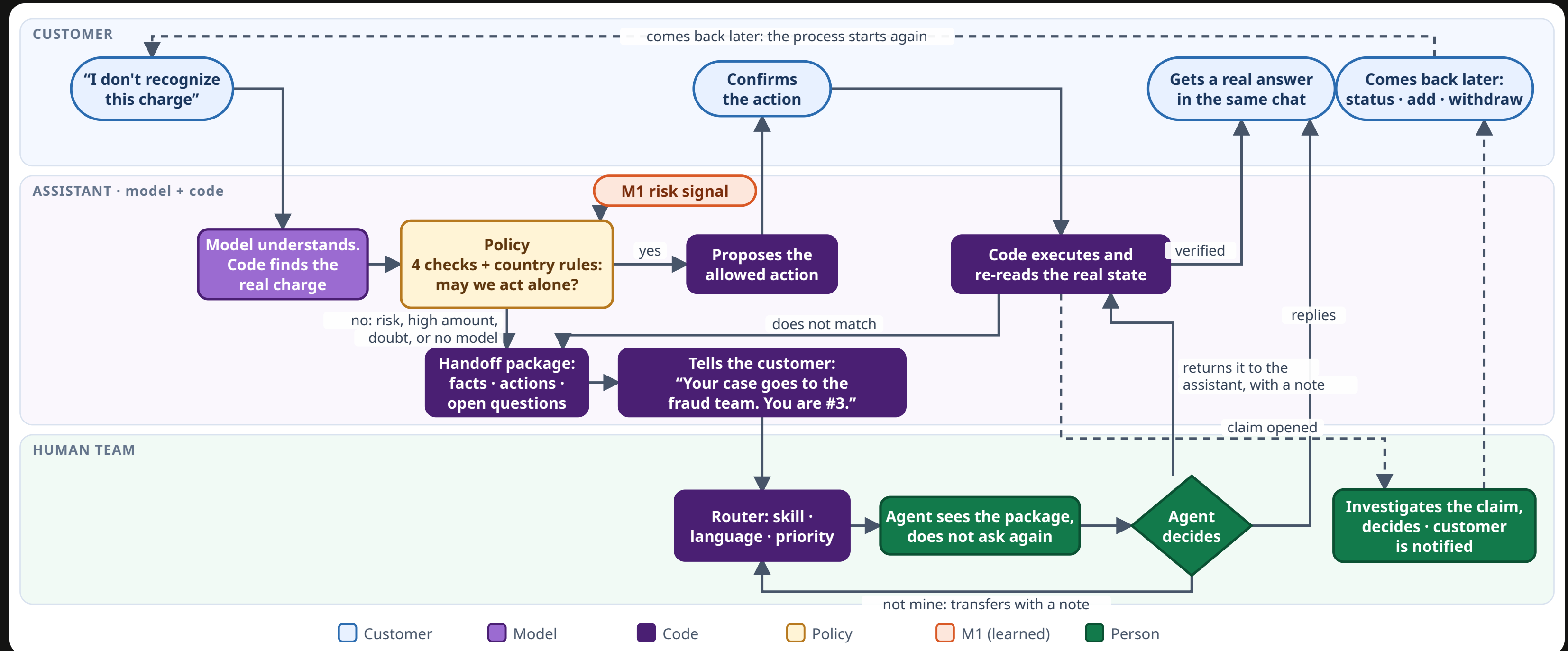
For the human agent

A complete case

For the bank

Every action with its rule and evidence

The model understands. The code decides and acts.



The model **cannot execute** any banking action.

Lora reports an action as done **only after the database confirms it.**

If the model fails, **no further action runs**: a person takes the case.

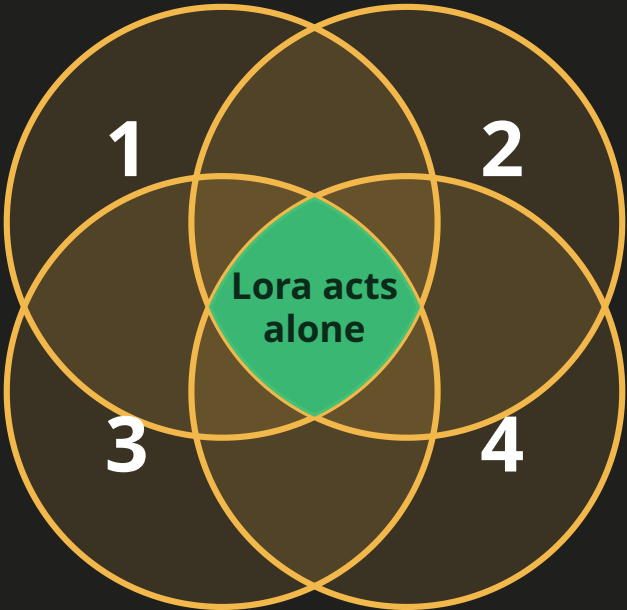
Row-Level Security: each customer reads only their own rows

Four policy checks control actions. One learned signal informs them.

THE POLICY · FOUR CHECKS, ALL MUST PASS

1 · Irreversibility against certainty

2 · Same case, same treatment



3 · A person always available

4 · Care for the relationship

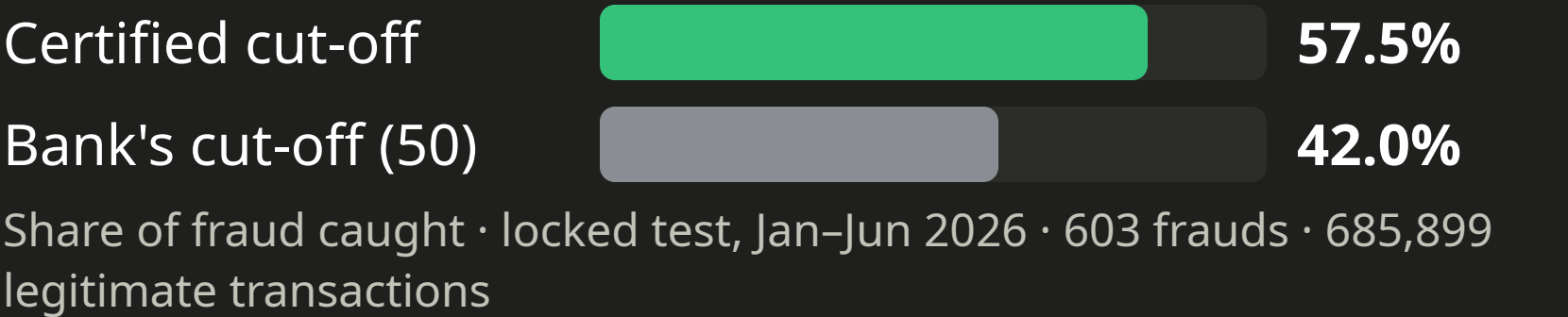
The intersection defines which actions Lora may perform without a person. Conceptual view: the four conditions are evaluated in code.

The database re-read · after she acts

The text check · before Lora speaks

THE FRAUD SIGNAL · CERTIFIED RULE: SCORE > 30

The bank already provides a fraud score for most transactions. We calibrated it and **certified** a cut-off: above it, Lora may offer a card block.



	Recommended blocks	Wrong, observed	95% upper bound (FDR)
Certified	347	0	0.86%
Bank's own	253	0	1.18%

Learn 2 years

Certify 6 months

Test once  6 months

23.5 M rows · 13 tables

Bronze · raw → Silver · contracts + quality checks → Gold · demo subset

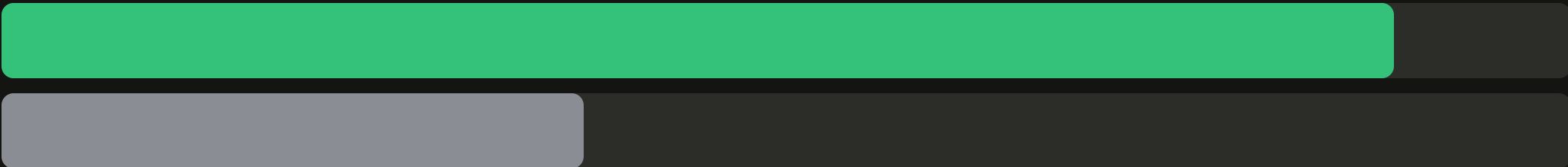
In our test, rules outside the prompt improved safety

Final run on unseen cases

62 cases (12 in Portuguese) · normal · ambiguous · needs a person · attacks · tool failures · scored from the database, not from the text

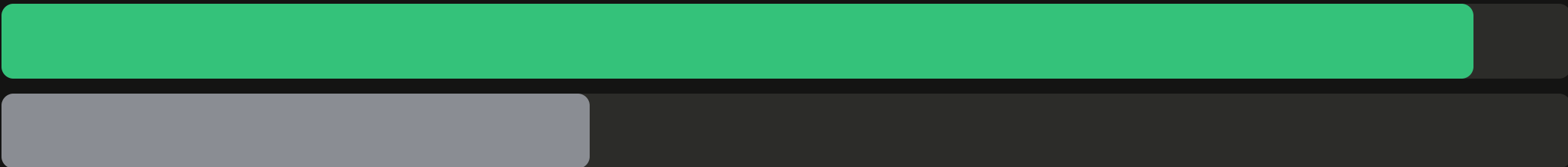
Cases passed

all expected outcomes met,
nothing unsafe



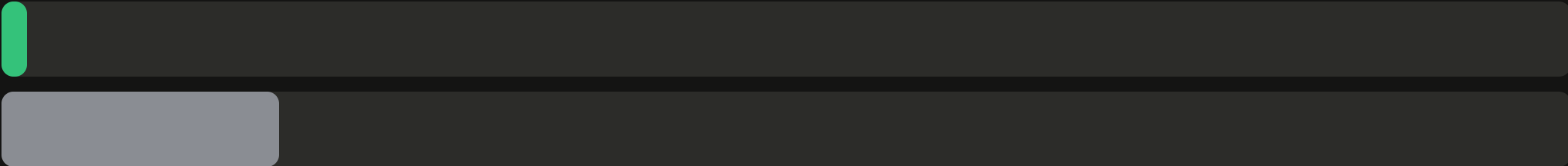
Correct escalation

of the 16 cases that needed a
person



Unsafe outcomes

lower is better



 Lora  plain assistant: same model, data, identity and tools; rules only in the prompt

Latency p50 / p95 per turn: 2.5 / 4.8 s · cost per case ≈ 0.0017 USD at public model prices · bars on one 0–100% scale · brackets: 95% Wilson interval

Same model and tools. Different control architecture. Different results in this test.

Offline results on synthetic data, one pass on 62 cases never used in development: “tested in our battery”, not “validated”. Lora's unsafe outcome (F6) is described in the report. Measured on version d466792; later changes are listed in the report.

What it would take to make it real

BUILT

- Traceability for every conversation turn and model call
- Bounded retries · safe fallback to a person
- Skills-based routing on the dataset's synthetic agent roster
- Supervisor, audit and back-office screens
- Recent-charge selection with “None of these” (tested; not yet evaluated with the live model)

Synthetic data, not production

TESTED

- Prompt injection
- Tool failures and model failures
- Row-level security and identity
- Spanish and Portuguese cases (Portuguese wording checked by back-translation)
- Policy tested across 98,304 combinations of input facts
- 275 automated tests

Small evaluation: 62 cases

NEXT

- Real bank systems integration
- Native review of the Portuguese
- A larger evaluation
- Evaluate paid model capacity and latency
- Deposits and disbursements (today: a person)

Portuguese checked by back-translation, not by a native reviewer

The model understands. The code decides and acts. Every action is verified.



Lora · Daniel Pineda

Factored AI & Data Hackathon 2026 · repository and live demo: see the links in the submission